

PROJECT DOCUMENTATION

Project Title	Automated Network Request Management in ServiceNow
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Project Type	Individual / Solo Project
Platform	ServiceNow Personal Developer Instance (PDI)
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This document records the table, automation, policy, and catalog configuration used to manage Network Requests through the ServiceNow Service Catalog and Flow Designer implementation.

1. Custom Table Configuration

Activity 1: Creation of Table

The foundation for the solution is the Network Database custom table. The following figures show the initial table creation and its basic configuration.

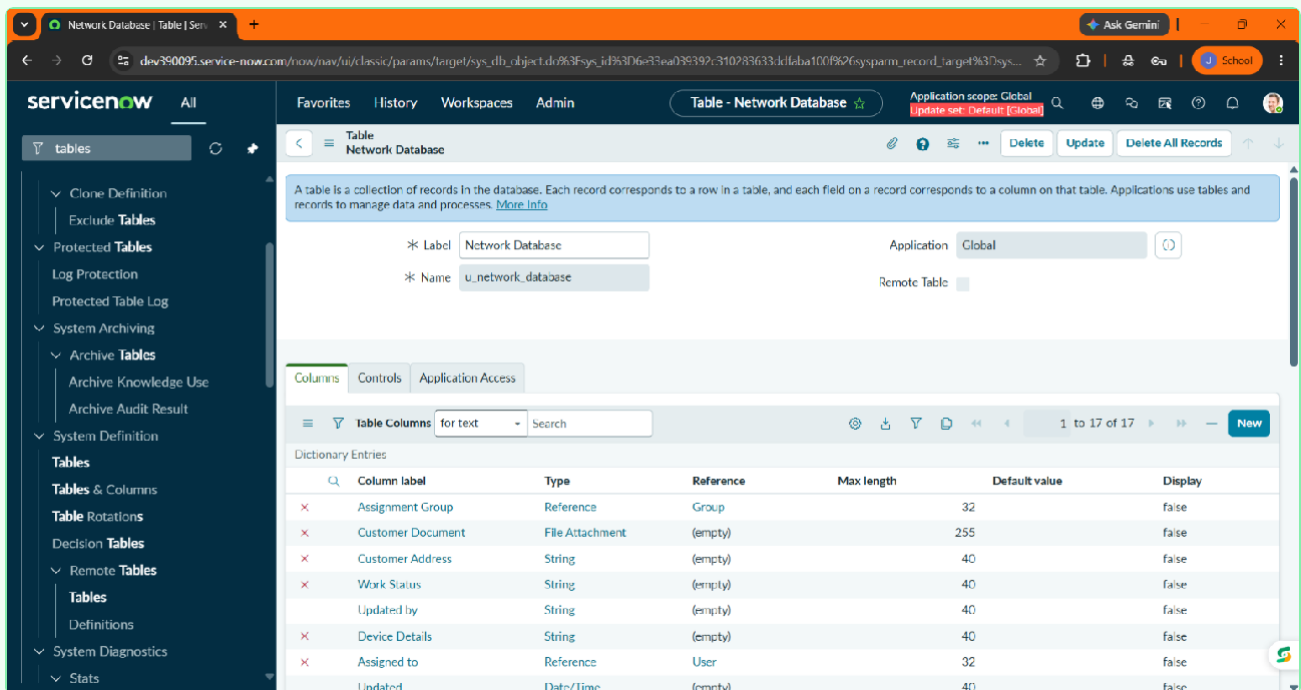


Figure 1. Network Database custom table.

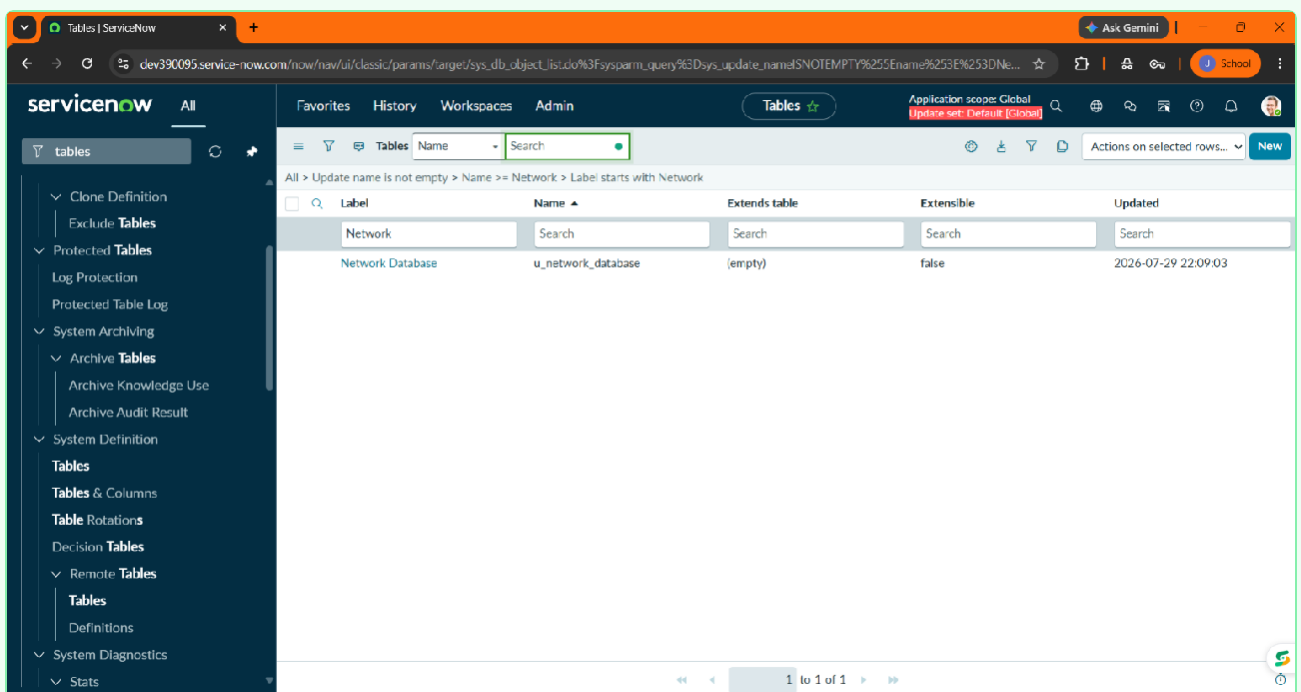


Figure 2. Network Database table configuration.

Activity 2 & 3: Creation of Fields and Defining Properties

We define the necessary fields required for the request records, applying the correct data types, max lengths, labels, and access controls for security.

Access Controls (4)							
Security Data Filters							
Labels (1)							
Database Indexes (3)							
Table Subscription Configuration (1)							
Access Controls							
☐	Name	Decision Type	Operation	Type	Active	Updated by	Updated
☐	u_network_database	Allow If	delete	record	true	admin	2026-07-29 22:09:04
☐	u_network_database	Allow If	write	record	true	admin	2026-07-29 22:09:04
☐	u_network_database	Allow If	read	record	true	admin	2026-07-29 22:09:04
☐	u_network_database	Allow If	create	record	true	admin	2026-07-29 22:09:04

Figure 3. Columns and field properties.

Table - Network Database							
Table Columns							
Dictionary Entries							
☐	Column label	Type	Reference	Max length	Default value	Display	
☒	Assignment Group	Reference	Group	32		false	
☒	Customer Document	File Attachment	(empty)	255		false	
☒	Customer Address	String	(empty)	40		false	
☒	Work Status	String	(empty)	40		false	
☒	Updated by	String	(empty)	40		false	
☒	Device Details	String	(empty)	40		false	
☒	Assigned to	Reference	User	32		false	
☒	Updated	Date/Time	(empty)	40		false	
☒	Type Of Devices	Choice	(empty)	255		false	
☒	Sys ID	Sys ID (GUID)	(empty)	32		false	
☒	Created by	String	(empty)	40		false	
☒	Request For	String	(empty)	40		false	
☒	Email	String	(empty)	40		false	
☒	Request Number	String	(empty)	40		false	
☒	Updates	Integer	(empty)	40		false	
☒	Created	Date/Time	(empty)	40		false	
☒	Date Of Enquiry	Date	(empty)	255		false	
☐	Insert a new row...						

Figure 4. Access controls for Network Database.

2. Flow Designer Automation

Activity 1 & 2: Creation of Flow and Configuring Trigger

The flow automates the lifecycle of a request. It begins by triggering upon catalog submission and retrieving the necessary variables provided by the requester.

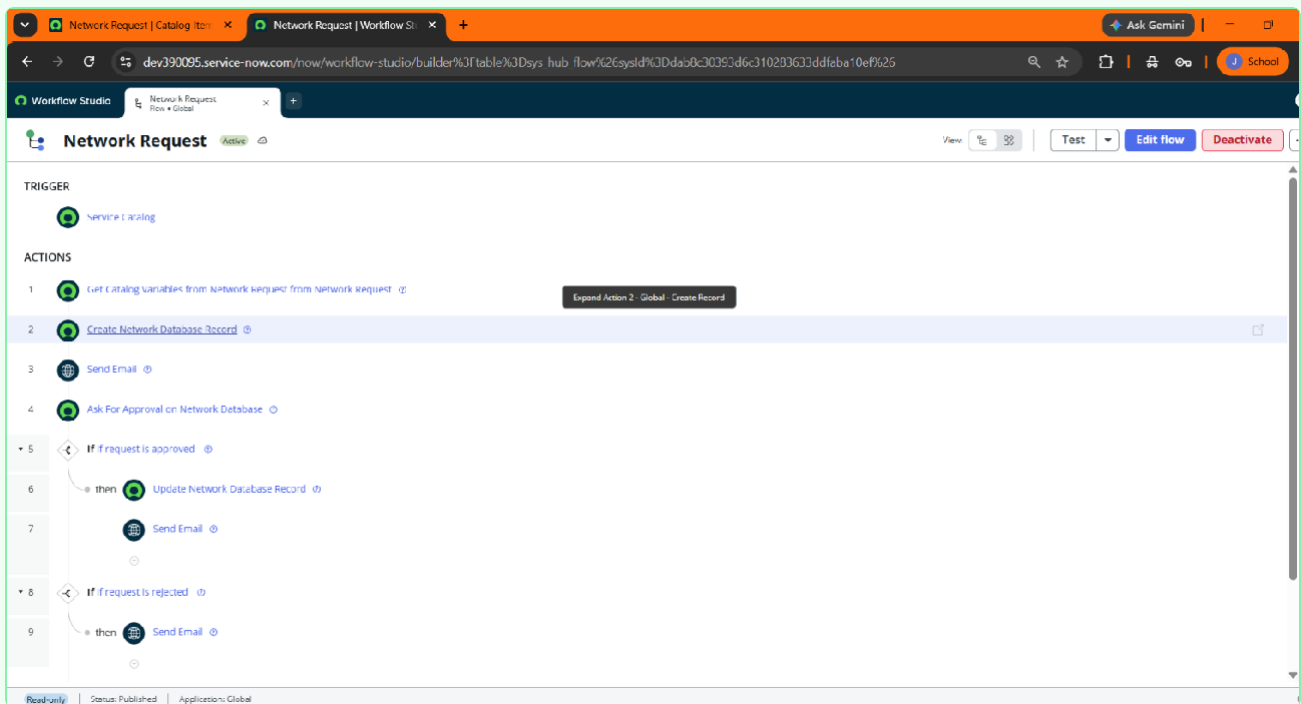


Figure 5. Network Request flow chart in Flow Designer.

The screenshot shows the configuration for the 'Get Catalog Variables' action. The action is selected from the dropdown. The input is 'Submitted Request (Requested Item)'. The template catalog item is 'Network Request'. The 'Available' list is empty, and the 'Selected' list contains the following variables:

- on_the_behalf_of
- email
- phone_number
- user_name
- proof_of_document
- request_for
- is_this_a_new_connection
- if_this_is_a_relocation_plea
- please_provide_address_he
- type_of_devices

A note at the bottom states: 'Note: If removing a variable from the 'Selected' list, it will be moved to 'Available' list only if the variable is from the selected Template Catalog Items and Variable Sets. Otherwise, the variable is removed from both 'Available' and 'Selected' lists.'

Figure 6. Get Catalog Variables action.

Activity 3: Configuring Actions - Approval

An approval action is implemented followed by a decision condition to dynamically route the workflow based on whether the request is approved or rejected.



Figure 7. Ask for Approval action.

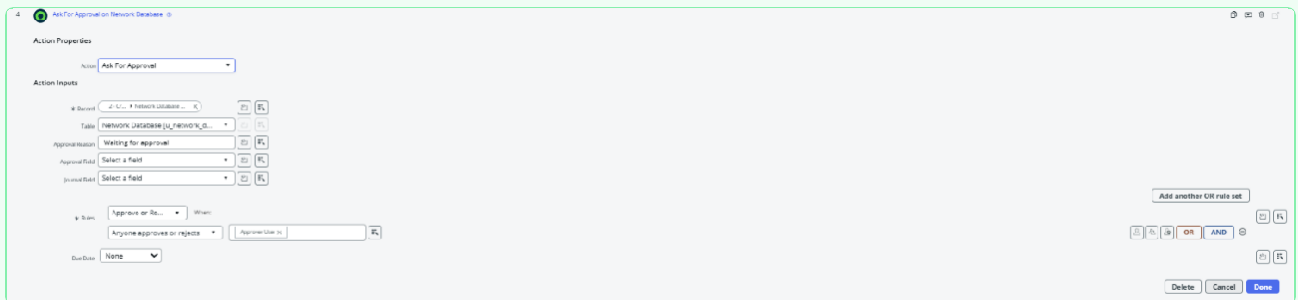


Figure 8. Approved decision condition.

Approved Request Path

For approved requests, the flow updates the current request status, creates a new entry in the Network Database using the catalog variables, and sends a confirmation email.

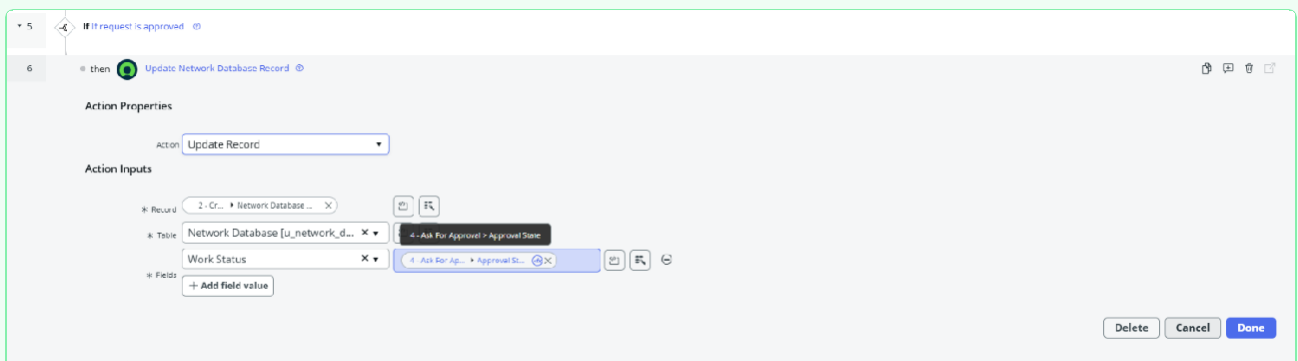


Figure 9. Approved-path Update Record action.

2 **Create Network Database Record**

Action Properties

Action: **Create Record**

Action Inputs

Table: **Request Number** **Request For** **Work Status** **Assignment Group** **Date Of Enquiry** **Device Details**

Fields: **Customer Address** **Type Of Devices** **Customer Document** **Email** **Assigned to**

Mapping details:

- Request Number: Trigger - Service... → Num...
- Request For: 1 - Get Catalog Va... → Na...
- Work Status: New
- Assignment Group: Network
- Date Of Enquiry: 1 - Get Catalog ... → Creat...
- Device Details: 1 - Get Catalo... → type of dev...
- Customer Address: 1 - Ge... → provide device det...
→ If this is a relocation pleas...
- Type Of Devices: 1 - Get Catalo... → type of dev...
- Customer Document: 1 - Get Cat... → proof of docu...
- Email: 1 - Get Catalog Va... → Em...
- Assigned to: 1 - Get Catalog V... → request...

+ Add field value

Figure 10. Create Record action and field mapping.

3 **Send Email**

Action Properties

Action: **Send Email**

Action Inputs

Target Record: 2 - Cr... → Network Database...

Table: Network Database [a_network_d...

Include Watermark: ☒

To: Trigger - Service C... → Em...

CC:

BCC:

Subject: Request has been created

Body:

Hello! 1 - Get Catalog Va... → Em...

We have been received your request! 2 - Crd... → Request No...

sorry for the inconvenience and your request will resolved in 2 business working days.

Thankyou.

Figure 11. Send Email action for approval.

Figure 12. Approval email received by the requester.

Rejected Request Path

If the request is rejected, the flow records this outcome, updates the request record accordingly, and sends a notification email explaining the rejection.

Figure 13. Rejected decision condition.

Figure 14. Rejected-path Update Record action.

9 then Send Email

Action Properties

Action: Send Email

Action Inputs

Target Record: 2 - Cr... Network Database ...

Table: Network Database [u_network_d...

Include Watermark: ☒

* To: 1 - Get Catalog Variabl... em...

CC:

BCC:

* Subject: Request is rejected

Body: Request is rejected

Delete Cancel Done

Figure 15. Rejection email action.

3. Catalog UI Policy and Client Script

To ensure data accuracy and guided form behaviour, Catalog UI Policies and a Client Script are applied. These dictate field visibility and mandatory requirements dynamically.

Variables (12)

Variable Sets (1)

Catalog UI Policies (3)

Catalog Client Scripts

Available For

Not Available For

Categories (1)

Catalogs (1)

Catalog Data Lookup Definitions

Related Articles

Related Catalog Items

Assigned Topics

≡

Order

Search

🔍

🔗

🔍

📄

—

Actions on selected rows...

New

Catalog Item = Network Request

☐	Short description	Variable set	Conditions	Reverse if false	On load	Inherit	Updated	Order
☐	Type of device is others	(empty)		true	true	false	2025-07-30 01:22:21	100
☐	Location Details	(empty)		true	true	false	2025-07-30 06:14:47	100
☐	Relocation Details	(empty)		true	true	false	2025-07-30 06:17:32	100

1

to 3 of 3

Figure 16. Network Request Catalog UI policies.

Variables (5)

Catalog UI Policies

Catalog Client Scripts (1)

Included in (1)

Catalog Data Lookup Definitions

≡

▼

for text

Search

🕒

📄

🔍

📄

—

Actions on selected rows...

New

Variable set = Request information

☐

🔍

Name

Active

Table

View

Type

Updated

Requester details

true

onChange

2026-07-30 00:19:56

⏪

◀

1

to 1 of 1

▶

⏩

Figure 17. Catalog UI policy variable actions.


```

1 function onChange(control, oldValue, newValue, isLoading) {
2   if (isLoading || !newValue) {
3     g_form.setValue('email', '');
4     g_form.setValue('user_name', '');
5     g_form.setValue('phone_number', '');
6     return;
7   }
8
9   var gr = g_form.getReference('on_the_behalf_of', function(user) {
10    g_form.setValue('email', user.email);
11    g_form.setValue('user_name', user.user_name);
12
13    // Use Mobile Phone first, otherwise fall back to Business Phone
14    var userPhone = user.mobile_phone || user.phone || '';
15    g_form.setValue('phone_number', userPhone);
16  });
17 }

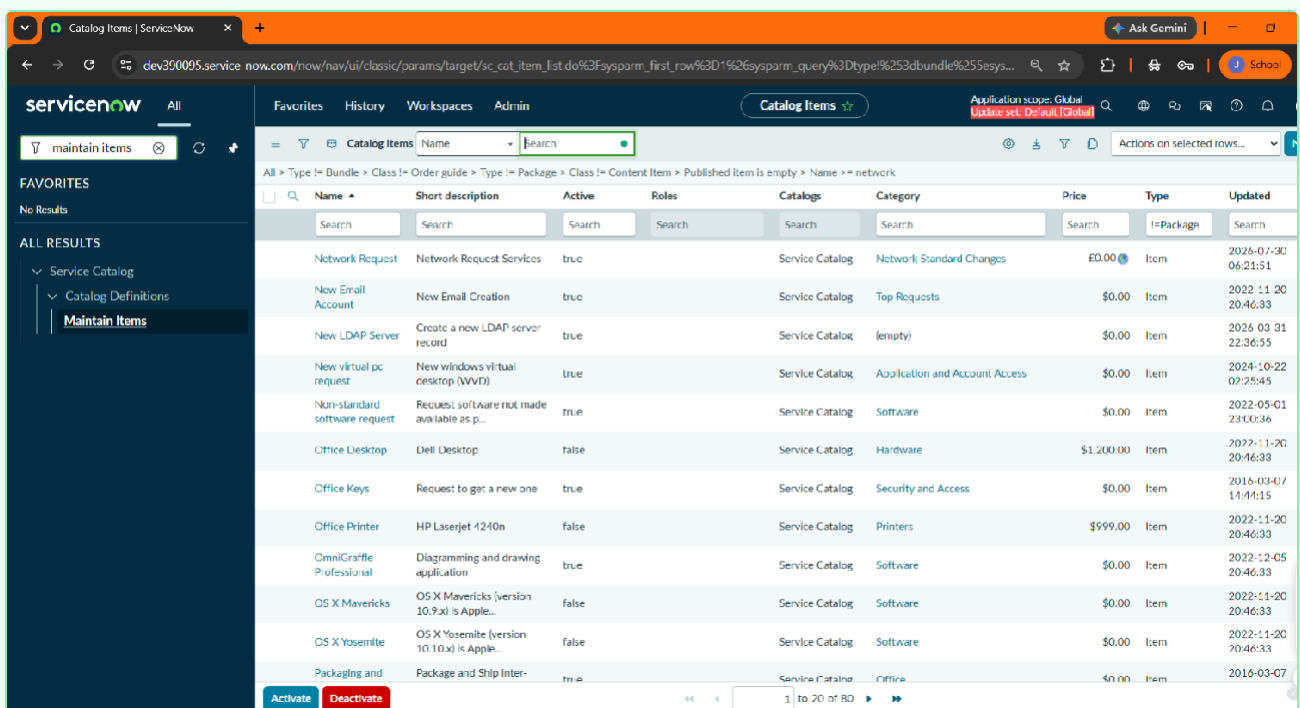
```

Figure 18. Catalog Client Script.

4. Service Catalog Configuration

Activity 1: Creation of Service Catalog Item

The Service Catalog item is created within Maintain Items, categorized correctly, and configured for user visibility.



Name	Short description	Active	Roles	Catalogs	Category	Price	Type	Updated
Network Request	Network Request Services	true		Service Catalog	Network Standard Changes	£0.00	Item	2026-07-30 06:21:51
New Email Account	New Email Creation	true		Service Catalog	Top Requests	\$0.00	Item	2022-11-20 20:46:33
New LDAP Server	Create a new LDAP server record	true		Service Catalog	(empty)	\$0.00	Item	2026-03-31 22:36:55
New virtual pc request	New windows virtual desktop (VVD)	true		Service Catalog	Application and Account Access	\$0.00	Item	2024-10-22 07:25:45
Non-standard software request	Request software not made available as p...	true		Service Catalog	Software	\$0.00	Item	2022-05-01 23:00:36
Office Desktop	Dell Desktop	false		Service Catalog	Hardware	\$1,200.00	Item	2022-11-20 20:46:33
Office Keys	Request to get a new one	true		Service Catalog	Security and Access	\$0.00	Item	2016-03-07 14:44:15
Office Printer	HP Laserjet 4240n	false		Service Catalog	Printers	\$999.00	Item	2022-11-20 20:46:33
OmniGraffle Professional	Diagramming and drawing application	true		Service Catalog	Software	\$0.00	Item	2022-12-05 20:46:33
OS X Mavericks	OS X Mavericks (version 10.9.x) is Apple...	false		Service Catalog	Software	\$0.00	Item	2022-11-20 20:46:33
OS X Yosemite	OS X Yosemite (version 10.10.x) is Apple...	false		Service Catalog	Software	\$0.00	Item	2022-11-20 20:46:33
Packaging and	Package and Ship Inter-	true		Service Catalog	Office	\$0.00	Item	2016-03-07

Figure 19. Maintain Items list.

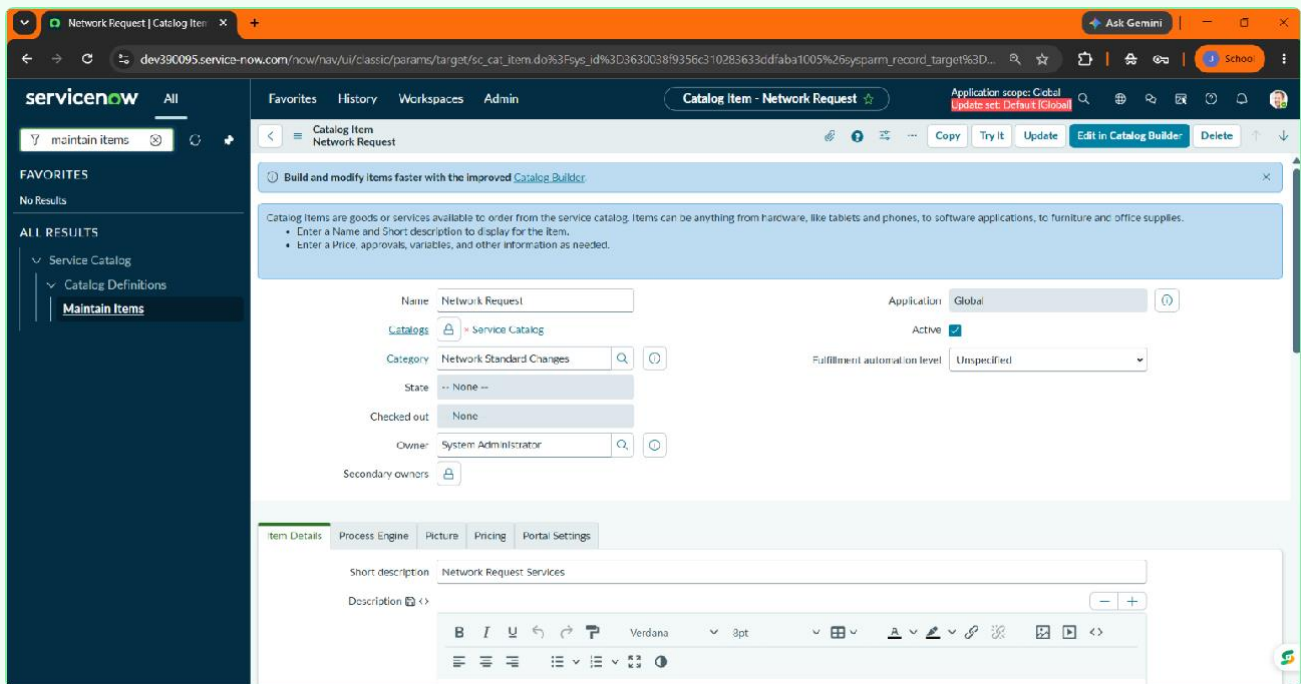


Figure 20. Network Request catalog item.

Activity 2 & 3: Variables Configuration and Types

The necessary questions are defined as catalog variables with specific types, ordering, and question text to gather the appropriate data from the user.

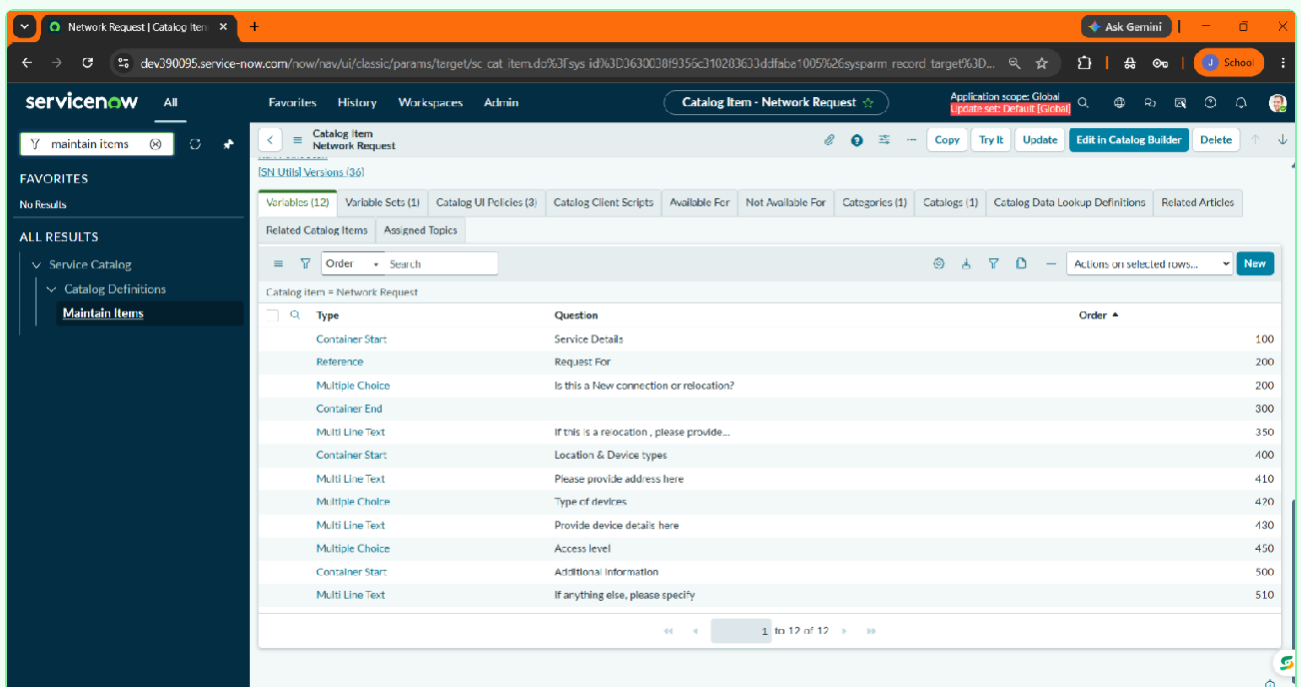


Figure 21. Catalog variables configured for Network Request.

☐	Name	Type	Question	Order
☒	on_the_behalf_of	Reference	on the behalf of	
☐	email	Single Line Text	email	
☐	phone_number	Single Line Text	phone number	
☐	user_name	Single Line Text	user name	
☐	proof_of_document	Attachment	proof of document	

Figure 22. Variable types and order.

Activity 4: Variable Set Configuration

A reusable variable set is created for consistency and attached to the main catalog item.

☐	Variable set	Order
☐	Request information	100

Figure 23. Network Request variable set.

Variable Set: Request Information

* Title: Request Information

* Internal name: request_information

Order: 100

Type: Single Row

Description:

Application: Global

Display title: ☒

Layout: 2 Columns Wide, one side, then the other

Figure 24. Variable set attached to the catalog item.

5. Catalog Submission & Validation

Activity 1 & 2: Search, Open, and Submit

Users can search for the catalog item, fill in the required fields guided by the UI policies, and submit their network request. A summary screen confirms the action.

servicenow Knowledge Catalog Requests System Status Cart Tours System Administrator

Home > Service Catalog > Standard Changes > Network Standard Changes > Network Request

Search Catalog

Network Request

Network Request Services

Indicates required

Request information

on the behalf of:

email:

phone number:

user name:

proof of document:

Request for:

Is this a New service item or an existing?

☒ New

☐ Relocation

*Please provide address here

Quantities:

Delivery Times: 2 Days

Figure 25. Search results showing the Network Request catalog item.

servicenow Knowledge Catalog Requests System Status Cart Tours System Administrator

Home > Search

network request

Sources

[All](#)

[Knowledge Bases](#)

[Catalogs](#)

FILTERS

Knowledge Bases

Category

Author

Last modified

View Count

Catalogs

Category

All results for "network request"

Network Request

Network Request Services

Request for Backup

Backup system, data files or database

Apple iPhone 13

Request for Apple iPhone 13

\$799.00

Apple iPhone 13 pro

Request for Apple iPhone 13 pro

\$999.00 + \$45.00 Monthly

<https://dev29005.servicenow.com/spaces/catalogItems?id=10d155ec0b1122e6305c0300a345082>

Figure 26. Network Request catalog form - requester information.

servicenow Knowledge Catalog Requests System Status Cart Tours System Administrator

Home > Request Summary - REQ0010009

Submitted : 2026-07-31 09:18:45
Request Number : REQ0010009
Requested for : Rachel Larison
Estimated Delivery : 2026-09-02

Item	Delivery Date	Stage	Price (each)	Quantity	Total
Network Request	2/05/18-02	▶ Assess or Scope Task	---	1	---

Total: \$0.00

Figure 27. Completed Network Request catalog form.

servicenow Knowledge Catalog Requests System Status Cart Tours System Administrator

Home > Service Catalog > Standard Changes > Network Standard Changes > Network Request

*Please provide address here
Gartner

Type of device
☐ Others
☒ Laptop
☐ Mobile

Provide device details here
Dell laptop

Access level
☐ High
☒ Low
☐ Medium

If anything else, please specify
Please solve as soon as possible

Add attachments
 Choose a file or drag it here.
 Copy and paste clipboard files here.

Quantity: 1
 Delivery Time: 2 Days
 Add to Cart
 Save as Draft
 Order Now

Figure 28. Submitted Network Request summary.

6. Flow Output Verification

Activity 3: Verify Approved Record

Following approval, the system administrator verifies that the Flow Designer successfully created the corresponding record in the Network Database table with all correct mappings.

servicenow All Favorites History Admin : Network Database - Created 2026-07-31 21:22:30 Search

Network Database
Created 2026-07-31 21:22:30 Update Delete

Request Number	RI Q0010009	Email	rachel.larrison@example.com
Request For	Rachel Larrison	Assigned to	Rachel Larrison
Work Status	approved	Date Of Enquiry	2012-02-17
Customer Address	Dell laptop	Customer Document	Click to add...
* Type Of Devices	Laptop	Assignment Group	Network
Device Details	laptop		

Update Delete

Figure 29. Network Database records created by the flow.

servicenow All Favorites History Workspaces Admin Network Databases Search

Network Databases Request Number Search Actions on selected rows... New

Request Number	Email	Device Details	Created	Date Of Enquiry
REQ0010008	amnic.approver@example.com	laptop	2026-07-30 07:32:25	2011-09-16
REQ0010006	alene.rabeck@example.com	laptop	2026-07-30 06:26:43	2012-02-17
REQ0010007	allen.schwartz@example.com	mobile	2026-07-30 06:41:11	2012-02-17
REQ0010005	alejandro.mascall@example.com	laptop	2026-07-30 06:07:01	2012-02-17
RI Q0010004	aileen.mottern@example.com	mobile	2026-07-30 05:52:49	2012-02-17
REQ0010009	rachel.larrison@example.com	laptop	2026-07-31 21:22:30	2012-02-17
RI Q0010003	adela.cervantes@example.com	mobile	2026-07-30 05:47:19	2012-02-17
REQ0010002	abraham.lincoln@example.com	laptop	2026-07-30 05:34:58	2013-07-23

1 to 8 of 8

Figure 30. Approved Network Database request record.